

Appendix 1

MATHEMATIC GROUNDS

This appendix is devoted to providing succinct mathematical grounds required in quantum image processing. For details on most of the ideas touched on here, you can consult basic probability theory (Ross, 2008), linear algebra (Friedberg, Insel, & Lawrence, 2003; Nielsen & Chuang, 2000), and wavelet theory (Ruch & Van Fleet, 2009).

Basic Probability Theory

Sample space is the set of all possible outcomes of an experiment, and is denoted by S . For instance, If the outcome of an experiment consists in the determination of the sex of a newborn child, then $S = \{g, b\}$, where the outcome g means that the child is a girl and b that it is a boy.

An event is any subset E of the sample space. If the outcome of the experiment is contained in E , then we say that E has occurred. For instance, if $E = \{g\}$, then E is the event that the child is a girl. For each event E of the sample space S , $n(E)$ is the number of times in the first n repetitions of the experiment that the event E occurs. Then the probability of the event E , $P(E)$ is defined as

$$P(E) = \lim_{n \rightarrow \infty} \frac{n(E)}{n}. \quad (1)$$

The basic notion of probability theory is that of a random variable. A random variable is a real-valued function defined on the sample space, which is determined by the outcome of the experiment. A random variable that can take on at most a countable number of possible values is said to be discrete. For a discrete random variable X , we define the probability mass

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function $p(a)$ of X by $p(a)=p(X=a)$. In this book we shall only be concerned with random variables which take their value from a finite set of values, and we always assume that this is the case. That is, if X must assume one of the values $x_1, x_2, \dots, x_n$, then

$$\sum_{i=1}^n p(x_i) = 1, \quad (2)$$

where $p(x_i) \geq 0$.

The expression of a random variable X , denoted by $E[X]$, is defined by

$$E(X) = \sum_i x p(x). \quad (3)$$

Therefore, the expected value of X is a weighted average of the possible values that X can take on, each value being weighted by the probability that X assumes it.

The variance of a random variable X is defined by

$$Var(X) = E[(X - E(X))^2] = E(X^2) - E(X)^2. \quad (4)$$

The square root of the $Var(X)$ is called the standard deviation of X , and we denote it by $SD(X)$. That is,

$$SD(X) = \sqrt{Var(X)}. \quad (5)$$

Suppose that an experiment, whose outcome can be classified as either a success or a failure is performed, a random variable X is said to be a Bernoulli random variable, if its probability mass function is given by

$$\begin{cases} p(0) = p(X = 0) = p, \\ p(1) = p(X = 1) = 1 - p, \end{cases} \quad (6)$$

where $X=1$ when the outcome is a success and $X=0$ when it is a failure.

The random variables X and Y are said to be independent if, for any two sets of real numbers A and B ,

$$P(x \in A, y \in B) = P(x \in A)P(y \in B). \quad (7)$$

Theorem A.1 (The strong law of large numbers) Let $X_1, X_2, \dots$ be a sequence of independent and identically distributed random variables, each having a finite mean $\mu = E(X_i)$. Then, with probability 1,

$$\lim_{n \rightarrow +\infty} \frac{X_1 + X_2 + \dots + X_n}{n} = \mu. \quad (8)$$

Linear Algebra

Linear algebra is the study of vector spaces and of linear operations on those vector spaces. A good understanding of quantum mechanics is based upon a solid grasp of elementary linear algebra. Some definitions and basic operations in linear algebra are defined as follows.

Definition A.2 (Group) A group is a set G , together with a map of $G \times G$ into G , denoted as $g * h$, with the following properties:

1. Associativity: For all $g, h, k \in G$, $g * (h * k) = (g * h) * k$.
2. There exists an identity element e in G such that for all $g \in G$, $g * e = e * g = g$.
3. For all $g \in G$, there exists $h \in G$, $g * h = h * g = e$. That is, h is the inverse of g , denoted as g^{-1} .

If $g * h = h * g$ for all $g, h \in G$, then the group is said to be commutative.

Definition A.3 (Field) A field $\langle \mathbb{F}, +, \cdot \rangle$ is a set $\square$ with two operations, addition and multiplication, such that

1. $\langle \mathbb{F}, + \rangle$ is a commutative group under addition, and its identity element is denoted as 0.
2. Let $\mathbb{F}^*$ be a set $\square$ without the element 0, then $\langle \mathbb{F}^*, \cdot \rangle$ is a commutative group under multiplication, and its identity element is denoted as 1.
3. Distributivity: For all $a, b, c \in \mathbb{F}$, $a(b+c) = ab+ac$.

Definition A.4 (Vector Space) A vector space over $\square$ is a set V endowed with two operations: vector addition $V \times V \rightarrow V$, denoted by $(v, w) \rightarrow v + w$, and

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scalar multiplication $\mathbb{F} \times V \rightarrow V$, denoted by $(a,v) \rightarrow av$, satisfying the following properties:

1. V is a commutative group under vector addition.
2. 2. Scalar multiplication satisfies the following identities:

$$\begin{cases} a(bv) = (ab)v, \text{ for all } v \in V \text{ and } a, b \in \mathbb{F} , \\ 1v = v, \text{ for all } v \in V. \end{cases} \quad (9)$$

3. 3. Scalar multiplication and vector addition are related by the following distributive laws:

$$\begin{cases} (a + b)v = av + bv, \text{ for all } v \in V \text{ and } a, b \in \mathbb{F} , \\ a(v + w) = av + aw, \text{ for all } v, w \in V \text{ and } a \in \mathbb{F} . \end{cases} \quad (10)$$

A subset $S \subseteq V$ is said to be linearly dependent if there exists a linear relation of the form $\sum_{i=1}^k a_i v_i = 0$, where $v_1, v_2, \dots, v_k$ are distinct elements of S and at least one of the coefficients a_i is nonzero; S is said to be linearly independent otherwise. A basis for V is a subset $S \subseteq V$ that is linearly independent and spans V . If V has a finite basis, then V is said to be finite-dimensional, and otherwise it is infinite-dimensional.

The vector space of most interest to us is $\mathbb{C}^n$, the space of all n -tuples of complex numbers, $(c_1, c_2, \dots, c_n)$. The elements of a vector space are called vectors, and we will sometimes use the column matrix notation

$$\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \quad (11)$$

to indicate a vector. There is an addition operation in $\mathbb{C}^n$ defined which takes pairs of vectors to other vectors, i.e.,

$$\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} + \begin{bmatrix} z_1 \\ \vdots \\ z_n \end{bmatrix} = \begin{bmatrix} c_1 + z_1 \\ \vdots \\ c_1 + z_n \end{bmatrix}. \quad (12)$$

where the addition operations on the right are just ordinary additions of complex numbers. The scalar multiplication in $\mathbb{C}^n$ is defined by

$$a \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} ac_1 \\ \vdots \\ ac_n \end{bmatrix}, \quad (13)$$

where a is a complex number, that is, $a \in \mathbb{C}$, and the multiplications on the right are ordinary multiplication of complex numbers.

In this book, we will use the standard notation of quantum mechanics for linear algebraic concepts. The standard quantum mechanical notation for a vector in a vector space is the symbol $|\psi\rangle$, where ψ is a label for the vector. The $|\cdot\rangle$ notation is used to indicate that the object is a vector.

Suppose $|v_1\rangle, |v_2\rangle, \dots, |v_n\rangle$ is a basis in $\mathbb{C}^n$, then any vector $|v\rangle$ in the vector space $\mathbb{C}^n$ can be written as a linear combination $|v\rangle = \sum_{i=1}^n a_i |v_i\rangle$ of vectors. For instance, a basis $|v_1\rangle, |v_2\rangle$ is a subset in $\mathbb{C}^2$ such that

$$|v_1\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, |v_2\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \quad (14)$$

Since any vector

$$|v\rangle = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \quad (15)$$

in $\mathbb{C}^2$ can be written as a linear combination $|v\rangle = a_1 |v_1\rangle + a_2 |v_2\rangle$ of the vectors $|v_1\rangle$ and $|v_2\rangle$, we say that the vectors $|v_1\rangle$ and $|v_2\rangle$ span the vector space $\mathbb{C}^2$.

Definition A.5 (Inner Product) Let V be a vector space, then a function $(\bullet, \bullet)$ from $V \times V$ to $\mathbb{C}$ is an inner product if it satisfies the requirements that:

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1. $(\bullet, \bullet)$ is linear in the second argument,

$$(|v\rangle, \sum_i \lambda_i |w_i\rangle) = \sum_i \lambda_i (|v\rangle, |w_i\rangle); \quad (16)$$

2. $(|v\rangle, |w\rangle) = (|w\rangle, |v\rangle)^*$, where $(\bullet)^*$ denotes complex conjugate of a complex number;
3. $(|v\rangle, |v\rangle) \geq 0$ with equality if and only if $|v\rangle = 0$.

For instance, an inner product in $\mathbb{C}^n$ is defined by

$$((v_1, \dots, v_n), (w_1, \dots, w_n)) = \begin{bmatrix} v_1^* & \dots & v_n^* \end{bmatrix} \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix}. \quad (17)$$

Definition A.6 (Inner Product Space) An inner product space is a vector space equipped with an inner product.

Definition A.7 (Hilbert Space) In the finite-dimensional complex vector spaces that come up in quantum computation and quantum information, a Hilbert space is exactly the same thing as an inner product space.

Wavelet Theory

Here we introduce the discrete Haar wavelet transform, the discrete Daubechies D4 wavelet transform, and the discrete wavelet packet transform.

The Discrete Haar Wavelet Transform

Definition A.8 (The discrete Haar wavelet transform) Suppose that N is an even positive integer. We define the discrete Haar wavelet transform as

$$H_N = \begin{bmatrix} F_{N/2} \\ G_{N/2} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix}, \quad (18)$$

where the block $F_{N/2}$ is called the average block, and the block $G_{N/2}$ is called the detail block.

Suppose V is a vector in the N -dimensional vector space $\mathbb{R}^N$ of real numbers, where N is a positive integer divisible by 2^n . We compute the discrete Haar wavelet transform of V ,

$$y^1 = H_N V = \begin{bmatrix} y^{1,a} \\ y^{1,d} \end{bmatrix}, \quad (19)$$

where the vector $y^{1,a}$ represents the $N/2$ -length average portion of the transform, and $y^{1,d}$ denotes the $N/2$ -length detail portion of the transform. The second iteration of the discrete Haar wavelet transform is

$$y^2 = \begin{bmatrix} H_{N/2} y^{1,a} \\ y^{1,d} \end{bmatrix} = \begin{bmatrix} y^{2,a} \\ y^{2,d} \\ y^{1,d} \end{bmatrix}, \quad (20)$$

where the vectors $y^{2,a}$ and $y^{2,d}$ are the $N/4$ -length vectors that represent the average and detail portions, respectively, of the discrete Haar wavelet transformation applied to the vector $y^{1,a}$.

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Since N is divisible by 2^n , we can iterate a total of n times, and the j th iteration of the discrete Haar wavelet transform, where $j=1,2,\dots,n$, is

$$y^j = \begin{bmatrix} y^{j,a} \\ y^{j,d} \\ y^{j-1,d} \\ y^{j-2,d} \\ \vdots \\ y^{2,d} \\ y^{j1d} \end{bmatrix}, \quad (21)$$

where $y^{j,a} \in \mathbb{R}^{N/2^j}$, and $y^{k,d} \in \mathbb{R}^{N/2^k}$, and $k=1,2,\dots,j$.

We can also write the iterated discrete Haar wavelet transform in terms of products of block matrices. Let I_n and O_n denote the $n \times n$ identity and zero matrices, respectively. We denote $H_{N,0} = H_N$ and

$$H_{N,j} = \text{diag}[H_{N/2^j}, I_{N/2^j}, I_{N/2^{j-1}}, \dots, I_{N/4}, I_{N/2}] \quad (22)$$

for $j=1,2,\dots,n$, where $\text{diag}[C_1, C_2, \dots, C_n]$ is a matrix with blocks $C_1, C_2, \dots, C_n$ on the main diagonal and zeros elsewhere. Then, we give the following definition:

Definition A.9 (The j th iteration of the discrete Haar wavelet transform)

The j th iteration of the discrete Haar wavelet transform is expressed as $H_{N,j-1} \dots H_{N,1} H_{N,0}$, that is,

$$y^j = H_{N,j-1} \dots H_{N,1} H_{N,0} V. \quad (23)$$

Definition A.10 (The inverse of the j th iteration of the discrete Haar wavelet transform) Since $H_{N,j}^{-1} = H_{N,j}^T$, where $(\bullet)^{-1}$ and $(\bullet)^T$ are the inverse and transpose of matrix, we obtain the inverse of the iterated discrete Haar wavelet transform as $H_{N,0}^T H_{N,1}^T \dots H_{N,j-1}^T$, i.e.,

$$H_{N,0}^T H_{N,1}^T \dots H_{N,j-1}^T y^j = V. \quad (24)$$

The Discrete Daubechies D4 Wavelet Transform

Definition A.11 (The discrete Daubechies D4 wavelet transform) Suppose that N is an even positive integer. The discrete Daubechies D4 wavelet transform is defined as

$$D_N = \begin{bmatrix} h_0 & h_1 & h_2 & h_3 & 0 & 0 & \cdots & 0 & 0 & 0 & 0 \\ 0 & 0 & h_0 & h_1 & h_2 & h_3 & \cdots & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & 0 & \cdots & h_0 & h_1 & h_2 & h_3 \\ h_2 & h_3 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & h_0 & h_1 \\ h_3 & -h_2 & h_1 & -h_0 & 0 & 0 & \cdots & 0 & 0 & 0 & 0 \\ 0 & 0 & h_3 & -h_2 & h_1 & -h_0 & \cdots & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \cdots & h_3 & -h_2 & h_1 & -h_0 \\ h_1 & -h_0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & h_3 & -h_2 \end{bmatrix}, \quad (25)$$

where $h_0 = \frac{1+\sqrt{3}}{4\sqrt{2}}$, $h_1 = \frac{3+\sqrt{3}}{4\sqrt{2}}$, $h_2 = \frac{3-\sqrt{3}}{4\sqrt{2}}$ and $h_3 = \frac{1-\sqrt{3}}{4\sqrt{2}}$.

Proposition A.12 (The Matrix T_N is Orthogonal) The matrix D_N given in Definition A.11 is orthogonal, that is, $D_N D_N^T = I_N$.

Definition A.13 (The j th iteration of the discrete Daubechies D4 wavelet transform) The j th iteration of the discrete Daubechies D4 wavelet transform is expressed as,

$$D_{N,j-1} \cdots D_{N,1} D_{N,0}, \quad (26)$$

where $D_{N,0} = D_N$, and

$$D_{N,j} = \text{diag}[D_{N/2^j}, I_{N/2^j}, I_{N/2^{j-1}}, \dots, I_{N/4}, I_{N/2}].$$

Definition A.14 (The inverse of the j th iteration of the discrete Daubechies D4 wavelet transform) Since the matrix $D_{N,j}$ is orthogonal, we

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have $D_{N,j}^{-1} = D_{N,j}^T$. The inverse of the iterated discrete Daubechies D4 wavelet transform is $D_{N,0}^T D_{N,1}^T \cdots D_{N,j-1}^T$.

The Discrete Wavelet Packet Transform

The only difference in performing i iterations of the discrete wavelet packet transform and i iterations of the discrete wavelet transform is that we apply a wavelet transformation matrix to both the averages and details portion at each packet iteration step.

Let W_N be a discrete wavelet transform, for instance, $W_N = H_N$ and $W_N = D_N$ are the discrete Haar wavelet transform and the discrete Daubechies D4 wavelet transform, respectively. We have the following definition.

Definition A.15 (The j th iteration of the one-dimensional discrete wavelet packet transform) The j th iteration of the one-dimensional discrete wavelet packet transform is defined as

$$W_N^j = W_{N,j-1} W_{N,j-2} \cdots W_{N,1} W_{N,0}, \quad (27)$$

where $W_{N,0} = W_N$ and

$$W_{N,j} = \text{Diag}(W_{N/2^j}, W_{N/2^j}, \dots, W_{N/2^j})$$

is a matrix with 2^j blocks of $W_{N/2^j}$ on the main diagonal and zeros elsewhere.

Definition A.16 (The inverse of the j th iteration of the one-dimensional discrete wavelet packet transform) The inverse of the j th iteration of the one-dimensional discrete wavelet packet transform is defined as

$$(W_N^j)^{-1} = W_{N,0}^T W_{N,1}^T \cdots W_{N,j-1}^T. \quad (28)$$

Suppose that A is an $M \times N$ matrix where M, N are even positive integers. If we denote the columns of A by $a^1, a^2, \dots, a^N$, then the matrix product $W_M A$ can be written

$$W_M A = W_M [a^1, a^2, \dots, a^N] = [W_M a^1, W_M a^2, \dots, W_M a^N], \quad (29)$$

and the product $W_M A$ simply applies the discrete wavelet packet transform to each column of A . In practice, both columns and rows of a digital grayscale image are processed. That is, the two-dimensional discrete wavelet packet transform of $M \times N$ matrix A where both M, N are even positive integers is defined as $W_M A W_N^T$. Therefore, we give the following definition.

Definition A.17 (The j th iteration of the two-dimensional discrete wavelet packet transform) The j th iteration of the two-dimensional discrete wavelet packet transform is defined as

$$W_M^j A (W_N^j)^T, \quad (30)$$

where W_M^j and $(W_N^j)^T$ are the j th iteration of the one-dimensional discrete wavelet packet transform and its inverse.

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